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Elia Pizzati

Curriculum Vitae

NHFP Einstein Fellow at the Center for Astrophysics | Harvard&Smithsonian.
Interested in supermassive black holes, quasars, galaxy formation, cosmology, gravitational waves, cosmological simulations, empirical models, Bayesian methods.

19 papers, of which 7 as a first-author; >800 total citations (*h-index*: 13), of which >300 first-author (*first-author h-index*: 7). A complete publication list is enclosed and available on NASA/ADS.

Education

2021–2025 **PhD in Astrophysics**, *Leiden Observatory*, Leiden
Supervisors: Prof. Joe Hennawi and Prof. Joop Schaye.
Thesis: *From a biased perspective: quasars, mergers, and planet-forming discs* (here)

2019–2021 **Master's Degree in Physics**, *University of Pisa*, Pisa
Supervisors: Prof. Andrea Ferrara and Dr. Andrea Pallottini.
Final grade: *110/110 cum Laude*, with a special mention from the committee (*Abbraccio accademico*) for the "remarkable grades and exceptional thesis work". GPA: *30.0/30*

2016–2022 **Diploma in Physics**, *Scuola Normale Superiore*, Pisa
Best Italian University for Physics; admits on merit only about ten physics students per year out of several hundred applications. Final Grade: 100/100. GPA: *28.5/30*

2016–2019 **Bachelor's Degree in Physics**, *University of Pisa*, Pisa
Final grade: *110/110 cum Laude.* GPA: *29.7/30*

Research Appointments and Activities

2025-2028 **NHFP Einstein Fellow**, *Center for Astrophysics | Harvard&Smithsonian*

2021-2025 **PhD project**, *Leiden Observatory*

2020 **INFN/LIGO Exchange Program**, *IGC, Pennsylvania State University*

2019 **LEAPS Program**, *Leiden Observatory*

Achievements

2025 **NHFP Einstein Fellowship**

2022 **"Geppina Coppola" Prize for the best Master's Thesis in Astrophysics**

2022 **"Carlo Azeglio Ciampi" Prize for the best Italian Master's Thesis**

2019 **NSF/INFN Exchange Funds**

2019 **LEAPS Scholarship**

2016 **Scuola Normale Superiore Admission Test**

2016 **Prize scholarship "Meggiato-Moresco-Zotti-Savella-Carleo"**

2015-2016 **National Physics Olympiad (*bronze*) and National Astronomy Olympiad (*gold*)**

2015–2016 **National Philosophical Debate Tournament (*best orator*)**

Teaching and mentoring experience

Student supervision (Leiden University and UC Santa Barbara)
Boyi Ding (Master's student at Leiden, 2023-2024; primary supervisor, 1 publication)

2021-2025 **Teaching Assistant** (Leiden University; ≈300 hours per year)
Teaching the courses "Galaxies and Cosmology", "Galaxies: Structure and Dynamics", "Large Scale Structure" for Bachelor's and Master's students in the Astronomy department.

Service and Outreach

2024- **Public outreach**
Delivered public outreach talks at events such as Astronomy on Tap and in Italian schools, communicating astronomy to diverse audiences

2024- **Reviewer for MNRAS, PASJ**

2022-2023 **Borrel Committee at Leiden Observatory**
Organizer of weekly social gatherings (*borrels*) for colleagues in Leiden.

2020-2022 **Tutoring and teaching at SNS**
Tutor for Physics students at my university; contributed to holding a lecture within the "SNS Internship in Physics for High School Students".

2006–2020 **Scout and volunteering experience**
With over 10 years of experience in the Scout Association and as a children's entertainer, I have extensive experience managing large groups and delivering engaging activities for 100+ children; I also designed and led immersive astronomy programs for teenagers.

Recent/Upcoming Talks

Conference Contributions

Jul. 2026 **Settling the Dust: Obscured AGN in Galaxy Evolution**, *Lorentz Center*, Leiden, The Netherlands

Jun. 2026 **Supermassive Black Holes and Blue Notes**, *Université de Montréal*, Montreal, Canada

Apr. 2026 **Probing the Genesis of Supermassive Black Holes**, *Kavli IPMU*, copenhagen

Mar. 2026 **Climbing Cosmic Peaks**, *Sexten Center for Astrophysics*, Sexten, Italy

Nov. 2024 **Probing the Genesis of Supermassive Black Holes**, *Kavli IPMU*, Tokyo, Japan

Aug. 2024 **Cosmic Dawn Revealed by JWST**, *KITP*, Santa Barbara CA, USA

Jul. 2024 **The Origin and Evolution of Supermassive Black Holes**, Sexten, Italy

Jun. 2024 **Unveiling Black Hole Growth across Cosmic Time**, EAS Padova, Italy

Jun. 2024 **From Galaxies to Quasars and back**, EAS Padova, Italy

May 2024 **First Stars VII**, *Simons Foundation*, NYC, USA

Apr. 2024 **Massive Black Holes in the First Billion Years**, Kinsale, Ireland

Nov. 2023 **NOVA Fall School**, *ASTRON*, Dwingeloo, The Netherlands

Jul. 2023 **Reionization in the summer**, *MPIA*, Heidelberg, Germany

Jun. 2023 **First Light Conference**, *MIT*, Cambridge MA, USA

Nov. 2022 **Life-cycle of AGN**, ESA Madrid, Spain

Seminars and Colloquia

Jun. 2026 **Gemini North Talk**, *Noirlab*, Hilo HI, USA

May 2026 **HEAD seminar**, *Center for Astrophysics|Harvard&Smithsonian*, Cambridge MA, USA

Apr. 2026 **ITC Luncheon**, *Center for Astrophysics|Harvard&Smithsonian*, Cambridge MA, USA

Mar. 2026 **Galaxies Seminar**, *University of Michigan*, Ann Arbor MI, USA

Oct. 2025 **Ph.D. Colloquium**, *University of Leiden*, Leiden, The Netherlands

Sep. 2024 **Cosmology Seminar**, *UC Berkeley*, Berkeley CA, USA

May 2024 **Galaxy Formation Meeting**, *Center for Computational Astrophysics*, NYC, USA

May 2024 **Cosmology Talk**, *Princeton University*, Princeton NJ, USA

May 2024 **Monday Afternoon Talks**, *MIT*, Cambridge MA, USA

Dec. 2023 **Cosmo-Talk**, *Scuola Normale Superiore (SNS)*, Pisa, Italy

Public Talks

Dec. 2023 **Cerimonia dei Diplomi**, *Scuola Normale Superiore (SNS)*, Pisa, Italy

Nov. 2022 **Una stella del cielo**, *Osservatorio Astronomico di Capodimonte*, Napoli, Italy

Press

Articles

Oct. 2024 *Astronomen vinden oude eenzame quasars met onduidelijke oorsprong (in Dutch)*, covered in NOVA, Leiden Science, etc.

Oct. 2024 *Astronomers detect ancient lonely quasars with murky origins*, covered in MIT news, Phys.org, ScienceDaily, etc.

Press

Oct. 2024 *Astronomen vinden oude eenzame quasars met onduidelijke oorsprong (in Dutch)*, covered in NOVA, Leiden Science, etc.

Oct. 2024 *Astronomers detect ancient lonely quasars with murky origins*, covered in MIT news, Phys.org, ScienceDaily, etc.

References (alphabetical order)

Anna-Christina Eilers, *MIT* — *eilers@mit.edu*

Andrea Ferrara, *Scuola Normale Superiore (SNS)* — *andrea.ferrara@sns.it*

Joe Hennawi, *UCSB and Leiden Observatory* — *joe@physics.ucsb.edu*

Andrea Pallottini, *Scuola Normale Superiore (SNS)* — *andrea.pallottini@sns.it*

Giovanni Rosotti, *University of Milano* — *giovanni.rosotti@unimi.it*

Bangalore Sathyaprakash, *Penn State University* — *bss25@psu.edu*

Matthieu Schaller, *Leiden Observatory* — *schaller@strw.leidenuniv.nl*

Joop Schaye, *Leiden Observatory* — *schaye@strw.leidenuniv.nl*

Publications

First author

1. **Elia Pizzati**, Joseph F Hennawi, Joop Schaye, Anna-Christina Eilers, Jiamu Huang, Jan-Torge Schindler, Feige Wang, *"Little Red Dots" cannot reside in the same dark matter halos as comparably luminous unobscured quasars*, Monthly Notices of the Royal Astronomical Society, Volume 539, Issue 4, June 2025, Pages 2910–2925, doi.org/10.1093/mnras/staf660
2. **Elia Pizzati**, Joseph F Hennawi, Joop Schaye, Matthieu Schaller, Anna-Christina Eilers, et al. (15 authors), *A unified model for the clustering of quasars and galaxies at $z \approx 6$*, Monthly Notices of the Royal Astronomical Society, Volume 534, Issue 4, November 2024, Pages 3155–3175, doi.org/10.1093/mnras/stae2307
3. **Elia Pizzati**, Joseph F Hennawi, Joop Schaye, Matthieu Schaller, *Revisiting the extreme clustering of $z \approx 4$ quasars with large volume cosmological simulations*, Monthly Notices of the Royal Astronomical Society, Volume 528, Issue 3, March 2024, Pages 4466–4489, doi.org/10.1093/mnras/stae329
4. **Elia Pizzati**, Giovanni P Rosotti, Benoît Tabone, *Constraining turbulence in protoplanetary discs using the gap contrast: an application to the DSHARP sample*, Monthly Notices of the Royal Astronomical Society, Volume 524, Issue 2, September 2023, Pages 3184–3200, doi.org/10.1093/mnras/stad2057
5. **E Pizzati**, A Ferrara, A Pallottini, L Sommovigo, M Kohandel, S Carniani, *[CII] Haloes in ALPINE galaxies: smoking-gun of galactic outflows?*, Monthly Notices of the Royal Astronomical Society, Volume 519, Issue 3, March 2023, Pages 4608–4621, doi.org/10.1093/mnras/stac3816

6. **Elia Pizzati**, Surabhi Sachdev, Anuradha Gupta, and Bangalore Sathyaprakash, *Toward inference of overlapping gravitational-wave signals*, Physical Review D, vol. 105, no. 10, 2022, doi.org/10.1103/PhysRevD.105.104016
7. **E Pizzati**, A Ferrara, A Pallottini, S Gallerani, L Vallini, D Decataldo, S Fujimoto, *Outflows and extended [CII] haloes in high-redshift galaxies*, Monthly Notices of the Royal Astronomical Society, Volume 495, Issue 1, June 2020, Pages 160–172, doi.org/10.1093/mnras/staa1163

Second / third author

1. J Huang, **E Pizzati**, J F Hennawi, J Schaye, M Schaller, B Snyder, Y Kang, *The Impact of Cosmic Variance and Satellites on JWST Clustering Measurements at Redshift around 6*, *submitted*, arXiv:2605.11077, doi:10.48550/arXiv.2605.11077
2. J Huang, J F Hennawi, **E Pizzati**, F Wang, J Yang, J B Champagne, et al. (24 authors), *Clustering of $z \sim 6.6$ Quasars and [O III] Emitters Constrains Host Halo Masses and Duty Cycles in 25 ASPIRE Fields*, *submitted*, arXiv:2602.04974, doi:10.48550/arXiv.2602.04974
3. B Ding[†], **E Pizzati**, J Schaye, J F Hennawi, W McDonald, M Schaller, *The luminosity function and clustering of bright quasars in the FLAMINGO cosmological simulations*, Monthly Notices of the Royal Astronomical Society, Volume 547, Issue 2, April 2026, stag308, doi.org/10.1093/mnras/stag308
4. S Onorato, J F Hennawi, **E Pizzati**, B P Venemans, A-C Eilers, *Homogeneous measurements of proximity zone sizes for 59 quasars in the Epoch of Reionization*, Monthly Notices of the Royal Astronomical Society, Volume 547, Issue 3, April 2026, stag388, doi.org/10.1093/mnras/stag388
5. A-C Eilers, R Mackenzie, **E Pizzati**, J Matthee, J F Hennawi, et al. (16 authors), *EIGER VI. The Correlation Function, Host Halo Mass and Duty Cycle of Luminous Quasars at $z \approx 6$*, The Astrophysical Journal, vol. 974, no. 275, 2024, doi.org/10.3847/1538-4357/ad778b

[†] Student-led publication (direct supervision).

Contributing author

1. X Shen, O Zier, A Smith, R Liu, R Kannan, T-E Bulichi, S M Koehler, V Springel, M Vogelsberger, L Hernquist, R P Naidu, A de Graaff, **E Pizzati**, D M Alexander, L C Ho, V Kokorev, G Leung, A-C Eilers, R C Hickox, *The Lumina Project: The Demographics of Active Galactic Nuclei from Quasars to Little Red Dots at $z \geq 3$*, *submitted*, arXiv:2605.24112, doi:10.48550/arXiv.2605.24112
2. T-E Bulichi, G C K Leung, A-C Eilers, P G Pérez-González, et al. (incl. **E Pizzati**; 19 authors), *MEOW: The increase in the obscured AGN fraction in mid-infrared from $0 \leq z \leq 6$ with JWST MIRI*, *submitted*, arXiv:2603.22393, doi:10.48550/arXiv.2603.22393
3. F Wang, J B Champagne, J Huang, J Yang, J F Hennawi, X Fan, et al. (incl. **E Pizzati**; 45 authors), *ASPIRE: The Environments and Dark Matter Halos of Luminous Quasars in the Epoch of Reionization*, *submitted*, arXiv:2602.04979, doi:10.48550/arXiv.2602.04979
4. J-T Schindler, J F Hennawi, F B Davies, S E I Bosman, F Wang, J Yang, A-C Eilers, X Fan, K Kakiichi, **E Pizzati**, R Nanni, *A first look at quasar–galaxy clustering at $z \simeq 7.3$*, Astronomy & Astrophysics, Volume 708, A160, April 2026, doi.org/10.1051/0004-6361/202557623
5. J-T Schindler, J F Hennawi, F B Davies, S E I Bosman, R Endsley, F Wang, J Yang, A J Barth, A-C Eilers, X Fan, K Kakiichi, M Maseda, **E Pizzati**, R Nanni, *A Little Red Dot at $z = 7.3$ within a Large Galaxy Overdensity*, Nature Astronomy, vol. 9, 1732–1744, 2025, doi.org/10.1038/s41550-025-02660-1
6. X Lin, F Wang, X Fan, Z Cai, et al. (incl. **E Pizzati**; 36 authors), *A SPectroscopic survey of biased halos In the Reionization Era (ASPIRE): Broad-line AGN at $z = 4$–5 revealed by JWST/NIRCam WFSS*, The Astrophysical Journal, vol. 974, no. 147, 2024, doi.org/10.3847/1538-4357/ad6565
7. L Sommovigo, A Ferrara, S Carniani, A Pallottini, P Dayal, **E Pizzati**, et al. (9 authors), *A new look at the infrared properties of $z \sim 5$ galaxies*, Monthly Notices of the Royal Astronomical Society, Volume 517, Issue 4, December 2022, Pages 5930–5941, doi.org/10.1093/mnras/stac2997

8. Y Fudamoto, R Smit, R A A Bowler, P A Oesch, et al. (incl. **E Pizzati**; 40 authors), *The ALMA REBELS Survey: Average [CII] 158 μm Sizes of Star-forming Galaxies from $z \approx 7$ to $z \approx 4$*, The Astrophysical Journal, vol. 934, no. 2, 2022, doi.org/10.3847/1538-4357/ac7a47
9. A Pallottini, A Ferrara, S Gallerani, C Behrens, M Kohandel, et al. (incl. **E Pizzati**; 13 authors), *A survey of high-z galaxies: SERRA simulations*, Monthly Notices of the Royal Astronomical Society, Volume 513, Issue 4, July 2022, Pages 5621–5641, doi.org/10.1093/mnras/stac1281